Consequence of Abel's Formula.

- Since $e^{-\int P(x)\,dx} \neq 0$ always, $W(x)$ and $W(x_0)$ have the same sign (or both zero).
- So it suffices to check W at a *single* point $x_0 \in I$.
- If $W(x_0) \neq 0$, then y_1, y_2 are independent throughout I.
- If $W(x_0) = 0$, then y_1, y_2 are dependent throughout I.
- This greatly simplifies verifying independence of solutions.
- Abel's formula also gives an explicit formula for W without solving the ODE fully.
- Unlike arbitrary functions, for solutions of a linear ODE, $W \equiv 0 \iff y_1, y_2$ are linearly dependent.
- This is a much stronger dichotomy than for general functions.



Example

- Verify $y_1 = e^{2x}, y_2 = e^{-3x}$ solve $y'' + y' - 6y = 0$ over $\mathbb{R}$ and find general solution.
- y_1 satisfies $(4 + 2 - 6)e^{2x} = 0$.
- y_2 satisfies $(9 - 3 - 6)e^{-3x} = 0$.
- $W(y_1, y_2) = y_1 y_2' - y_2 y_1' = -3e^{-x} - 2e^{-x} = -5e^{-x}$
- Since $W \neq 0$ for all $x \in \mathbb{R}$, $\{y_1, y_2\}$ is a linearly independent over $\mathbb{R}$.
- The GS is $\boxed{y = c_1 e^{2x} + c_2 e^{-3x}}$.



Example

- **Q.** Prove that $c_1e^{2x} + c_1xe^{2x}$ is the general solution $y'' - 4y' + 4y = 0$ over $\mathbb{R}$.
- Note that $y_1 = e^{2x}$ satisfies $(4 - 8 + 4)e^{2x} = 0$.
- Now $y_2' = e^{2x} + 2xe^{2x}$, $y_2'' = 4e^{2x} + 4xe^{2x}$
- Substituting y_2 in the LHS of differential eq, we get 0.
- To verify, linear independence over the domain, we need to check Wronskian must be different from zero over the domain.
- $W(y_1, y_2) = e^{2x}(e^{2x} + 2xe^{2x}) - xe^{2x}(2e^{2x}) = e^{4x} \neq 0$ for all $x \in \mathbb{R}$.
- Hence, the GS of the DE is $\boxed{y = c_1e^{2x} + c_2xe^{2x}}$ over $\mathbb{R}$.



Example

- **Q.** Prove that $c_1 x + c_2 x \ln x$ is the general solution of the diff. eq. $x^2 y'' - xy' + y = 0$, $x > 0$.
- Let $y_1 = x$. Then, $y_1' = 1, y_1'' = 0 \Rightarrow 0 - x + x = 0$. ✓
- Let $y_2 = x \ln x$. Then, $y_2' = \ln x + 1$, $y_2'' = 1/x$
- Substituting y_2: $x - x(\ln x + 1) + x \ln x = 0$. ✓
- To prove that the Wronskian is different from zero for all $x > 0$.
- Now $W(y_1, y_2) = x(\ln x + 1) - x \ln x = x \neq 0$ for $x > 0$.
- Hence, the GS of the DE is $\boxed{y = c_1 x + c_2 x \ln x}$.



Example

- **Q** If $xy'' - (x+1)y' + y = 0$ has $W(1) = e$, find $W(x)$ using Abel's formula.
- Standard form: $y'' - \dfrac{x+1}{x}y' + \dfrac{1}{x}y = 0 \;\Rightarrow\; P(x) = -\dfrac{x+1}{x}$
- From Abel formulation, we have $W' + PW = 0$, $W(x_0) = W_0$.
- Integrating it from x_0 to x, we will obtain

$$W(x) = W(x_0)e^{\int_{x_0}^{x} P(t)\,dt}$$

.

- Thus, $W(x) = W(1)\,e^{-\int_1^x P(t)\,dt} = e \cdot e^{\int_1^x \frac{t+1}{t}\,dt}$
- $\int_1^x \frac{t+1}{t}\,dt = (x + \ln x) - 1$

$$\boxed{W(x) = e \cdot e^{x-1+\ln x} = xe^x}$$



Existence and Uniqueness

- Theorem Consider $y'' + P(x)y' + Q(x)y = R(x)$, $x \in I$, $y(x_0) = y_0$, $y'(x_0) = y_0'$. If P, Q, R are continuous on an interval I containing x_0, then this IVP has a **unique** solution $y(x)$ valid on all of I. (Proof is not required).
- It underlies all subsequent theorems about the structure of solutions.
- Solution Space is a Vector Space: The set S of all solutions of $L[y] = y'' + P(x)y' + Q(x)y = 0$ on I forms a **vector space** over $\mathbb{R}$, under usual addition and scalar multiplication.
- This follows from the superposition principle and $L[0] = 0$.
- If P, Q are continuous on I, then $\dim(S) = 2$.
- Any two linearly independent solutions y_1, y_2 form a **basis** of S.
- This basis is called a **fundamental set of solutions**.



Example

- **Q** Find the largest interval guaranteeing a unique solution of $(x-1)y'' + 3y' + (\tan x)y = e^x,\ y(2) = 1, y'(2) = 0.$
- Writing the DE is standard form, we have

$$y'' + \frac{3y'}{x-1} + \frac{(\tan x)y}{x-1} = \frac{e^x}{x-1}.$$

- Standard form coefficients undefined at $x = 1$ and at $x = \frac{\pi}{2} + k\pi$.
- Nearest singularities to $x_0 = 2$: $x = \frac{\pi}{2} \approx 1.57$ (left), $x = \frac{3\pi}{2} \approx 4.71$ (right).

$$\boxed{\text{Largest interval} = \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)}$$



Fundamental Set of Solutions

- A pair $\{y_1, y_2\}$ of solutions of the homogeneous equation is a fundamental set if

  $W(y_1, y_2)(x) \neq 0$ for some (equivalently every) $x \in I$.
- Existence of a fundamental set is guaranteed by the existence-uniqueness theorem.
- Take y_1 solving $y(x_0) = 1, y'(x_0) = 0$, and y_2 solving $y(x_0) = 0, y'(x_0) = 1$.
- Then $W(y_1, y_2)(x_0) = 1 \neq 0$, so $\{y_1, y_2\}$ is fundamental.
- This proves existence of a basis for the solution space S.



Example

- **Q** Construct a fundamental set for $y'' + y = 0$ at $x_0 = 0$.
- We need two LI functions that satisfy the DE with addition to two initial conditions as prescribed earlier.
- y_1: $y_1(0) = 1, y_1'(0) = 0 \Rightarrow y_1 = \cos x$
- y_2: $y_2(0) = 0, y_2'(0) = 1 \Rightarrow y_2 = \sin x$
- $W(y_1, y_2)(0) = \cos 0 \cos 0 - \sin 0(-\sin 0) = 1 \neq 0$
- Hence $\{\cos x, \sin x\}$ is a fundamental set.

$$\boxed{y = c_1 \cos x + c_2 \sin x}$$



General Solution of Homogeneous Equation

- Theorem If $\{y_1, y_2\}$ is a fundamental set of solutions of $L[y] = 0$ on I, then every solution y of $L[y] = 0$ on I can be written as

  $y(x) = c_1 y_1(x) + c_2 y_2(x)$ for unique constants c_1, c_2.
- This expression is called the **general solution** (or complementary function) y_c. It represents *all* solutions of the homogeneous DE.
- A **particular solution** y_p of $L[y] = R(x)$ is any single function satisfying it (with no arbitrary constants).
- Theorem (Structure Theorem). If y_p is a particular solution of $L[y] = R(x)$, and $y_c = c_1 y_1 + c_2 y_2$ is the general solution of the associated homogeneous equation $L[y] = 0$, then the general solution of $L[y] = R(x)$ is
  $y(x) = y_c(x) + y_p(x) = c_1 y_1(x) + c_2 y_2(x) + y_p(x)$.



Proof Sketch of Structure Theorem

- If y is any solution of the non-homogeneous DE $L[y] = R(x)$, then define $u = y - y_p$.
- Since y_p is a particular solution of $L[y] = R(x)$, $L[y_p] = R(x)$.
- Then $L[u] = L[y] - L[y_p] = R(x) - R(x) = 0$.
- So u solves the homogeneous equation, hence $u = c_1 y_1 + c_2 y_2$.
- Therefore $y = y_p + c_1 y_1 + c_2 y_2$, proving every solution has this form.
- Conversely, $L[y_p + c_1 y_1 + c_2 y_2] = R(x) + 0 + 0 = R(x)$, so it is indeed a solution.
- This decomposition $(y_c + y_p)$ is fundamental to solving non-homogeneous equations.



Summary of Key Theorems

- Existence-Uniqueness: IVP has a unique solution on all of I when P, Q, R continuous.
- Superposition: linear combinations of homogeneous solutions are solutions.
- Solution space S of $L[y] = 0$ is a 2-dimensional vector space.
- Wronskian test + Abel's formula: determine independence via a single point check.
- General solution of $L[y] = 0$: $y_c = c_1 y_1 + c_2 y_2$ (fundamental set).
- General solution of $L[y] = R(x)$: $y = y_c + y_p$.



Reduction of Order

- **Motivation.** For $y'' + P(x)y' + Q(x)y = 0$, the general solution needs *two* independent solutions.
- Sometimes only *one* solution $y_1(x)$ is known (by inspection, series methods, etc.).
- **Reduction of order** is a method to construct a second, independent solution y_2 from y_1.
- Idea: seek y_2 in the form $y_2(x) = v(x)\, y_1(x)$ for an unknown function $v(x)$.
- This reduces the 2nd order ODE for y_2 to a *1st order* ODE for v'.
- Hence the name: the order of the problem is effectively reduced by one.
- The method also works for non-homogeneous equations (finding y_p from y_c).



Reduction of Order

- **Setup.** Let $y_1(x)$ be a known nontrivial solution of $y'' + P(x)y' + Q(x)y = 0$ on an interval I, with $y_1(x) \neq 0$ on I.
- Assume a second solution of the form $y = v(x)\, y_1(x)$.
- Then, $y' = v'y_1 + vy_1'$ and $y'' = v''y_1 + 2v'y_1' + vy_1''$
- Substitute y, y', y'' into the given ODE, we obtain
  $v''y_1 + 2v'y_1' + vy_1'' + P(x)(v'y_1 + vy_1') + Q(x)vy_1 = 0$
  $\implies y_1 v'' + \big(2y_1' + P(x)y_1\big)v' + \big(y_1'' + P(x)y_1' + Q(x)y_1\big)v = 0$
  $\implies y_1 v'' + \big(2y_1' + P(x)y_1\big)v' = 0$
  ($\because$ y_1 solves the homogeneous equation).
- The above DE has **no** v **term**.



Reduction of Order

- Let $w = v'$. The equation becomes **first order** in w:

  $y_1 w' + \left(2y_1' + P(x)y_1\right) w = 0$

- This is separable:

  $$\frac{w'}{w} = -\frac{2y_1'}{y_1} - P(x)$$

- Integrating:

  $$\ln|w| = -2\ln|y_1| - \int P(x)\,dx + C$$

  $$\Rightarrow \quad w(x) = \frac{e^{-\int P(x)\,dx}}{y_1^2(x)} \qquad \text{(taking constant} = 1\text{)}$$



Reduction of Order

- Since $w = v'$, integrate once more to get v:

$$v(x) = \int \frac{e^{-\int P(x)\,dx}}{y_1^2(x)}\,dx$$

- Hence the **second solution formula**:

$$\boxed{y_2(x) = y_1(x) \int \frac{e^{-\int P(x)\,dx}}{y_1^2(x)}\,dx}$$

- Now $W(y_1, y_2) = y_1^2 w = e^{-\int P(x)\,dx} \neq 0$ (Homework), so y_1, y_2 are independent.
- This matches **Abel's formula** for the Wronskian.
- General solution: $y = c_1 y_1(x) + c_2 y_2(x)$.

